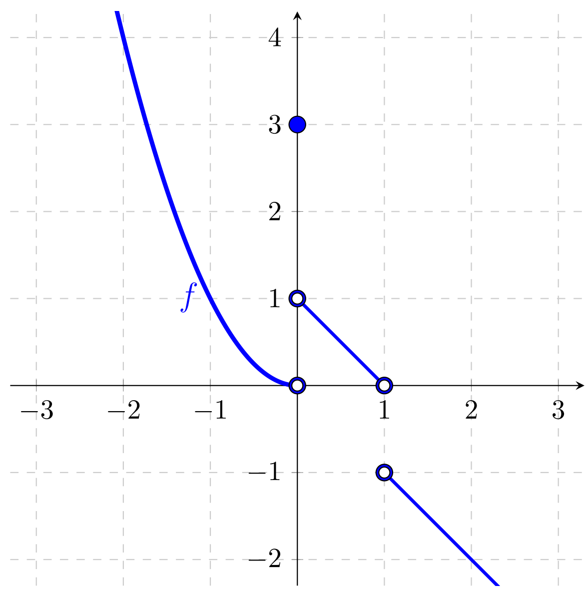
Problem 14

Problem 14

Problem 1 *We're given the following graph of a function:*



Use this graph to answer the following questions:

(a) *What is the domain of this function?*

Explanation. $(-\infty, 1) \cup (1, \infty)$

(b) *What is the range of this function?*

Explanation. $(-\infty, -1) \cup (0, \infty)$

(c) *What is the value of $f(0)$, $f(1)$, and $f(2)$?*

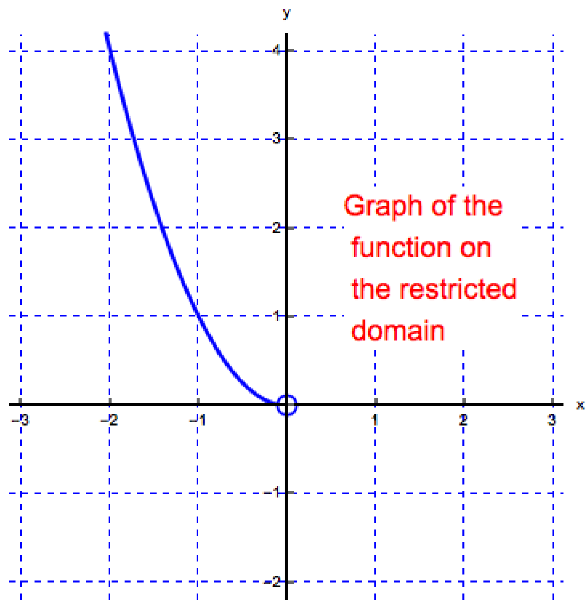
Explanation. $f(0) = 3$, $f(1)$ does not exist, $f(2) = -2$

(d) *Does this function have an inverse? (Why or why not?)*

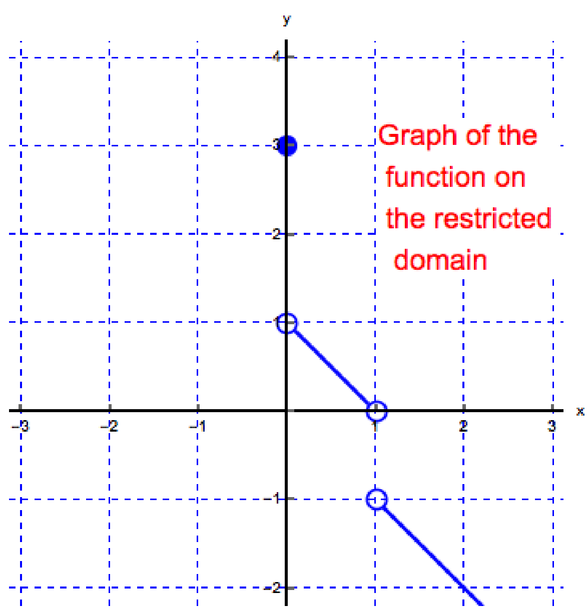
Explanation. No, the function does not have an inverse. It is not one-to-one (that is, it does not pass the horizontal line test).

(e) *Find at least two intervals on which the function is one-to-one.*

Explanation. The function becomes one-to-one when we restrict its domain to $(-\infty, 0)$:



The function also becomes one-to-one when we restrict its domain to $[0, 1) \cup (1, \infty)$:



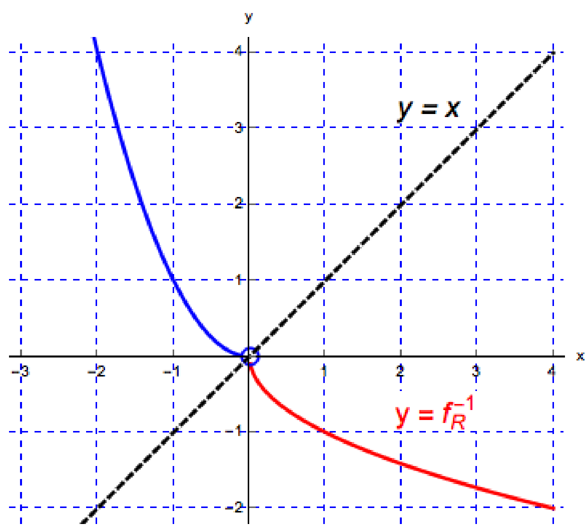
(f) *Find $f^{-1}(3)$ on a restricted domain of f.*

Explanation. In this case restrict the domain of f to $[0,1) \cup (1,\infty)$. By definition we have $f^{-1}(3) = y \iff 3 = f(y)$. Looking at the second graph of the restricted function we see that $f(0) = 3$, that is, $f^{-1}(3) = 0$.

(If we restrict the domain of f to $(-\infty, 0)$, then $f^{-1}(3) = y \iff 3 = f(y)$ implies that $f(-1.7) \approx 3$. Hence $-1.7 \approx f^{-1}(3)$.)

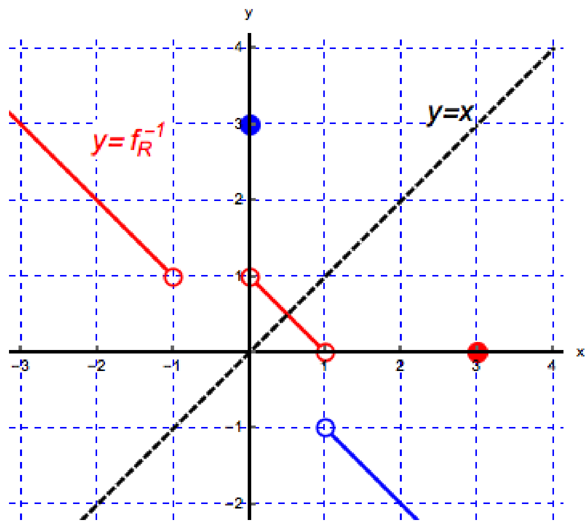
(g) *Using the restricted domain for f of $(-\infty, 0)$, sketch a graph of f^{-1}.*

Explanation. To graph the inverse, we can think of the graph being reflected over the line $y = x$. Another way to obtain the graph is to remember than if (x, y) is a point on the graph of f, then (y, x) is a point on the graph of f^{-1}. For example, $(-1, 1)$ is on the graph of f so $(1, -1)$ is on the graph of f^{-1}.



(h) *Using the restricted domain for f of $[0,1) \cup (1,\infty)$, sketch a graph of f^{-1}.*

Explanation. We can graph the inverse of f, when restricted to $[0,1) \cup (1,\infty)$, in pieces from left to right. Since $(0, 3)$ is a point on the graph of f, the point $(3, 0)$ in on the graph of f^{-1}. Then, we see on the graph of f, there is a linear piece going from $(0, 1)$ to $(1, 0)$. When we imagine reflecting this part of the graph of f over the line $y = x$, it reflects onto itself. The last piece of the graph of f is a line from $(1, -1)$ which appears to also contain the point $(-2, 2)$. Reflecting this part of f over the line $y = x$, we obtain a line starting at $(-1, 1)$ and through the point $(-2, 2)$.


y
x
$y = f_R^{-1}$
$y=x$
-3
-2
-1
1
2
3
4
-1
-2